

OFFICIAL TECHNICAL INSPECTION & CALIBRATION REPORT

AIR-GAPPED ON-PREMISE VERIFICATION ENCLAVE · PLANT ASSET DOSSIER

DOC ID: DOC-INS-2026-R204-098

DATE: 2026-10-10

CLEARANCE: INTERNAL / RESTRICTED

TARGET ASSET: REACTOR R-204 (Continuous Catalytic Hydrocracker)

1. Executive Summary & Vessel Integrity Assessment

A mandatory non-destructive ultrasonic and telemetry calibration inspection was completed for Reactor R-204. Primary vessel shell integrity meets ASME Section VIII Division 1 guidelines with nominal thickness at 7.8 mm (minimum allowable safety thickness: 6.5 mm). Operating baseline pressure is verified at 38.2 BAR. Maximum Allowable Working Pressure (MAWP) is strictly governed at 42.5 BAR with relief threshold at 43.0 BAR. No micro-fractures, abnormal acoustic emissions, or cavitation pitting were detected in the primary reaction loop.

2. Pressure Transducer & Relief Valve Calibration Matrix

Sensor Tag	Monitored Parameter	Observed Value	Baseline Standard	Variance	Status
PT-204A	Vessel Head Pressure	41.8 BAR	41.7 BAR	+0.1 BAR	PASS [VERIFIED]
PT-204B	Core Reaction Zone P	41.9 BAR	41.7 BAR	+0.2 BAR	PASS [VERIFIED]
TT-204	Vessel Skin Temperature	412.5 C	410.0 C	+2.5 C	PASS [NORMAL]
LT-204	Liquid Catalyst Level	68.4 %	68.0 %	+0.4 %	PASS [OPTIMAL]
PRV-204	Mechanical Relief Setpoint	42.5 BAR	42.5 BAR	0.0 BAR	PASS [GATED]
FT-204	Secondary Quench Flow	294 L/min	280 L/min	+14 L/min	PASS [ACTIVE]

3. Deterministic Safety Verification & Audit Proof

Lead Inspector: Devraj M., Senior Reliability Specialist (Badge #ENG-8842)
Verification Authority: FORGE Sovereign Policy Engine (Gate Action: PERMIT_READ)
Relief Valve PRV-204 Calibration: Verified with 0.0 BAR drift against NIST-traceable manometer.
Cryptographic Enclave Hash: SHA256: 7d8841a0e9b62f55c8290bc41de1fca3786214bf689392e6a

SOVEREIGN VERIFICATION SEAL
STATUS: DETERMINISTIC PASS
FAIL-CLOSED POLICY: ENFORCED
AIR-GAP ENCLAVE 01 · NO CLOUD
TIMESTAMP: 2026-10-10 01:20 IST

STANDARD OPERATING PROCEDURE & SAFETY PROTOCOLS

MANDATORY OPERATIONAL DIRECTIVES & MULTILINGUAL EMERGENCY INSTRUCTIONS

4. Standard Operating Procedure (SOP-R204-P)

PROTOCOL 1 [Cooling Loop Integrity]:

Maintain secondary quench flow strictly above 280 L/min during catalytic hydrocracking operations.
Any momentary drop below 260 L/min requires instantaneous activation of auxiliary pump AP-204B.

PROTOCOL 2 [Pressure Variance Boundary]:

Pressure divergence between dual redundant transmitters PT-204A and PT-204B must not exceed 0.5 BAR.
If variance exceeds 1.5 BAR for longer than 30 seconds, automated control must flag a sensor anomaly and escalate to human supervisor review before allowing any pneumatic control adjustments.

PROTOCOL 3 [Actuation & Relief Valve Security]:

Autonomous AI models are strictly prohibited from writing or triggering calibration offsets to valve PRV-204.
Physical setpoint modification requires cryptographic multi-factor authorization by the Plant Operations Lead.
Default policy enforcement is strictly FAIL-CLOSED.

5. Multilingual Emergency Directives / /

ENGLISH DIRECTIVE [OPERATIONAL LIMIT]:

CRITICAL SAFETY LIMIT: Do not exceed 42.5 BAR operating pressure.

Immediately isolate valve HV-204 and notify the shift supervisor if pressure exceeds 42.8 BAR.

HINDI DIRECTIVE [हिंदी आपातकालीन निर्देश]:

महत्वपूर्ण सुरक्षा सीमा: रिएक्टर परिचालन दबाव 42.5 BAR से अधिक न होने दें।
यदि दबाव 42.8 BAR से अधिक हो जाए, तो तुरंत वाल्व HV-204 बंद करें और शिफ्ट सुपरवाइजर को सूचित करें।

KANNADA DIRECTIVE [ಕನ್ನಡ ತುರ್ತು ನಿರ್ದೇಶನ]:

ನಿರ್ಣಾಯಕ ಸುರಕ್ಷತಾ ಮಿತಿ: ರಿಯಾಕ್ಟರ್ ಕಾರ್ಯಾಚರಣೆಯ ಒತ್ತಡವು 42.5 BAR ಮೀರುವುದನ್ನು ತಡೆಯಿರಿ.
ಒತ್ತಡವು 42.8 BAR ಮೀರಿದರೆ, ತಕ್ಷಣ ವಾಲ್ವ್ HV-204 ಅನ್ನು ಪ್ರತ್ಯೇಕಿಸಿ ಮತ್ತು ಪಾಳಿ ಮೇಲ್ವಿಚಾರಕರಿಗೆ ತಿಳಿಸಿ.

OFFICIAL PLANT CERTIFICATION & AIR-GAPPED AUDIT GUARANTEE

This multi-page technical report was rasterized and cryptographically verified on-premise without cloud transmission.